TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 4
Use of Multimedia



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 4 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 4, software, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Writing spaces** - what students write in every writing space
- **Unit 4 Test** - the marking scheme, textbook misprints and common mistakes

★ BEFORE THE FIRST LESSON

- ☐ GIMP, Vectorian Giotto, Audacity and Windows Movie Maker open on every computer.
- ☐ A microphone and speakers work with the teacher's computer.
- ☐ A shared folder has photographs of places of interest in Sri Lanka.

"Students remember what they create. Let them make, show and share."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Bits and colours	8
	B Work it out	6
	C RGB and HTML notation	8
	D Which one?	8
Chapter 2	A Complete the menu paths	8
	B Label the GIMP window	5
	C Which tool?	8
	D The Layers window	8
Chapter 3	A Name the frames	4
	B Frames and seconds	4
	C Label the Giotto window	8
	D Which tool?	8
	E Buttons and fills	8
Chapter 4	A Label the Audacity window	8
	B Audio or video software?	8
	C Which command or setting?	8
End	Unit 4 Test	40
TOTAL		**155**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Zoom into a photograph on the projector until the pixels show. Students never forget the squares.
- Let students work out 2 to the power 1 to 8 on the board before they fill the bits and colours table.
- Teach colours from bits as "which power of 2?": 256 = 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2, so 8 bits.
- Show 8f1c19 in GIMP's colour window, then change one digit and watch the colour change.
- Use one microphone and a quiet corner for recording. Let groups take turns.

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Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (30)	Ch 2 (29)	Ch 3 (32)	Ch 4 (24)	Test (40)	Total (155)
1							
2							
3							
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26							
27							
28							
29							

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Practicals at a Glance

Textbook activities on pages 102, 113 and 123

Practical	Students do	Check that ...
1 (p. 123, items 5 - 6)	zoom into a photograph; colour codes in GIMP's colour window; export a JPEG and a PNG file and compare them	they see square pixels; R 143, G 28, B 25 gives 8f1c19; the JPEG file is smaller and loses quality when zoomed (lossy), the PNG keeps it (lossless).
2 (p. 102)	a graphic from photographs, filters, graphics of places, an invitation, a banner	every picture is on its own layer; the layers are merged at the end; the .xcf is saved and a JPEG or PNG is exported; borders and the drop shadow are used.
3 (p. 113)	a motion text on their own image in Vectorian Giotto	the layers Background and Text; an effect on the text; a key frame at the end of the text on the Background layer; the .vgd is saved; the .swf plays in a browser.
4 (p. 123, items 1 - 4)	graphics, animations, a recorded and trimmed audio, a video clip about places in Sri Lanka	the audio is trimmed and exported (MP3); transitions, a title, captions, credits and music; the .wlmp project is saved and the movie saved For computer.

! SOFTWARE, SAFETY AND MATERIALS

- Install GIMP, Vectorian Giotto, Audacity and Windows Movie Maker before the unit (free software; the download addresses are in the textbook). Only you install software.
- A microphone (or headphones with a microphone) for Practical 4, and speakers for the class.
- A shared folder with photographs of places of interest in Sri Lanka (Sigiriya and others), so nobody needs unknown websites.
- Use only pictures and music that students made or are allowed to use. Students save often, with clear names.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Digital graphics	81 - 88	123 (items 5 - 6)
Chapter 2 - Graphic design with GIMP	89 - 102	102
Chapter 3 - Two-dimensional animation	102 - 113	113
Chapter 4 - Audio and video	113 - 123	123 (items 1 - 4)
Unit 4 in One Page	123 - 124 (summary)	-

★ TEACHING TIPS

- Remind students of hexadecimal from Grade 10 Unit 3: two hexadecimal digits hold one value from 0 to 255.
- Save one photograph as JPEG and as PNG and compare the file sizes in the folder: lossy is smaller.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What are the four basic elements of a digital graphic?	**Pixel, resolution, size, colour**
2. How many colours do 8 bits per pixel give?	**256**
3. Which colour model is used for printing?	**CMYK**
4. Name a lossless file format.	**GIF, PNG or RAW**
5. Which kind of graphic keeps its quality when enlarged?	**Vector graphic**
6. In which format does GIMP save its graphics?	**XCF**
7. Who makes the tween frames?	**The software**
8. What does File, Export Flash Movie make?	**A .swf file**
9. Which Audacity button removes the unselected parts?	**Trim Audio**
10. In which format is a Movie Maker project saved?	**.wlmp**

> PRACTICAL TEST - 40 MINUTES, 20 MARKS (OPTIONAL)

Use the computers of Practicals 2 - 4. Give the marks while the student shows the files.

Step	The student must ...	Marks
1	In GIMP make a new image of 640 × 400 px and fill it with a gradient (29c89c to e3216a).	3
2	Add the text **NATURE** (72 points, black) on its own layer; add a drop shadow.	4
3	Merge the visible layers, add a border, save the .xcf and export a JPEG.	3
4	In Vectorian Giotto put the JPEG on the work stage and make the text move with an effect.	4
5	Record a 5-second sentence in Audacity, trim it and export it as MP3.	3
6	Save the Giotto animation (.vgd) and export it (.swf); show it in a browser.	3
	Total	**20**

ANSWERS

Chapter 1

Digital graphics

A Bits and colours

8 marks

1. 1 bpp: 2^1 = **2**	**2.** 3 bpp: 2^3 = **8**
3. 16 colours = 2^4: **4 bpp**	**4.** 6 bpp: 2^6 = **64**
5. 256 colours = 2^8: **8 bpp**	**6.** 10 bpp: 2^{10} = **1,024**
7. 65,536 colours = 2^{16}: **16 bpp**	**8.** 24 bpp: 2^{24} = **16,777,216**

1 mark each. Colours = 2 to the power of the bits per pixel; for the bits, find the power of 2 that gives the colours. The textbook's "square root of the colours" (p. 83) works only for 16 colours - do not teach it.

B Work it out

6 marks

1. **Figure 4.4 (p. 84): 250 × 175 = 43,750 pixels**
2. **A new image in GIMP (p. 95): 1200 × 800 = 960,000 pixels; 960,000 × 24 ÷ 8 = 2,880,000 bytes**
3. **The graphic with text (p. 100): 640 × 400 = 256,000 pixels; 256,000 × 8 ÷ 8 = 256,000 bytes**
4. **The same graphic, 24 bits: 640 × 400 = 256,000 pixels; 256,000 × 24 ÷ 8 = 768,000 bytes**

1 mark for each empty cell (6). 8 bits = 1 byte. 768,000 bytes is 3 times 256,000 bytes: 24 bits per pixel need 3 times the space of 8 bits.

C RGB and HTML notation

8 marks

1. **Figure 4.9: R 143, G 28, B 25 → 8f1c19**
2. **Gradient colour 1 (p. 100): 29c89c → R 41, G 200, B 156**
3. **Gradient colour 2 (p. 100): e3216a → R 227, G 33, B 106**
4. **Black (p. 95): R 0, G 0, B 0 → 000000**

1 mark for each empty cell (8). Working: 143 = 8 × 16 + 15 = **8f**; 28 = 1 × 16 + 12 = **1c**; 25 = 1 × 16 + 9 = **19**. Back: 29 = 2 × 16 + 9 = 41, c8 = 12 × 16 + 8 = 200, 9c = 156; e3 = 227, 21 = 33, 6a = 106.

D Which one?

8 marks

It is ...	Lossy	Lossless	Raster	Vector
The file is as small as possible, but the quality is reduced.	☑	☐	☐	☐
It is made of an array of pixels in different colours.	☐	☐	☑	☐
It keeps a record of its lines, their start and end, and the colours.	☐	☐	☐	☑
The graphic opens again in its original form.	☐	☑	☐	☐
JPEG	☑	☐	☐	☐
Its quality is lost when its size changes.	☐	☐	☑	☐
GIF and PNG	☐	☑	☐	☐
It is made of a collection of straight or curved lines.	☐	☐	☐	☑

1 mark each. The textbook (p. 87) also lists TIFF and BMP as lossy; BMP is normally saved without compression, so the workbook asks only about JPEG.

TIP: Enlarge a raster and a vector star side by side on the projector. The difference teaches itself.

ANSWERS

Chapter 2

Graphic design with GIMP

A Complete the menu paths

8 marks

1. **Keep GIMP in one window: Windows → Single-Window Mode**
2. **Show or hide the tools of the toolbox: Edit → Preferences → Toolbox**
3. **Make the edge of a selection soft: Select → Feather**
4. **Paste a copied part as a new layer: Edit → Paste as → New Layer**
5. **Make a selection 2 pixels bigger: Select → Grow**
6. **Add a border to a graphic: Filters → Decor → Add Border**
7. **Add a shadow to text: Filters → Light and Shadow → Drop Shadow**
8. **Save the graphic as a JPEG or PNG file: File → Export**

1 mark each. Accept the path without arrows if the order is right (Filters, Decor, Add Border).

B Label the GIMP window

5 marks

1. Image Window	**2.** Brushes/Patterns/Gradients
3. The Main Toolbox	**4.** Layers, Channels, Paths
5. Tool Options	

1 mark each (textbook page 89).

C Which tool?

8 marks

1. Crop (Shift + C)	**2.** Fuzzy (Magic Wand) (U)
3. Bucket Fill (Shift + B)	**4.** Heal (H)
5. Scale (Shift + T)	**6.** Ellipse (E)
7. Colour Picker (O)	**8.** Blend (Gradient) (L)

1 mark each: the name and the key. Accept Fuzzy Select or Magic Wand, Blend or Gradient, Colour Picker or Eyedropper.

D The Layers window

8 marks

1. New	**2.** Group	**3.** Move	**4.** Duplicate layer
5. Anchor	**6.** Delete	**7.** Opacity	**8.** Eye (shows or hides the layer)

1 mark each (page 98). 3 is Move: the two arrows raise and lower the layer.

ANSWERS

Chapter 3

Two-dimensional animation

A Name the frames

4 marks

1. Tween frames	**2.** Blank key frame	**3.** Key frame	**4.** Frames

1 mark each. The grey frames 2 - 12 of the Text layer are "frames": they make the content last longer.

B Frames and seconds

4 marks

1. **24 - 2 = 22 tween frames**
2. **24 × 3 = 72 frames**
3. **24 ÷ 12 = 2 seconds**
4. **It becomes faster (twice as fast)**

C Label the Giotto window

8 marks

1. Geometrical Properties and Shapes	**2.** Built-in player	**3.** Work Stage	**4.** Drawing Toolbar
5. Properties Window	**6.** Timeline	**7.** Colours	**8.** Layers

1 mark each (page 104).

D Which tool?

8 marks

1. Sub selection tool (A)	**2.** Free transform tool (Q)
3. Ink bottle tool (S)	**4.** Line tool (N)
5. Insert bitmap (M)	**6.** Eyedropper tool (I)
7. Lasso tool (L)	**8.** Brush tool (B)

1 mark each: the name and the key (pages 105 - 108).

E Buttons and fills

8 marks

1. Rewind	**2.** End frame	**3.** Stop	**4.** Start
5. Radial	**6.** Solid	**7.** Bitmap	**8.** Linear

1 mark each. Linear and Radial are colour combinations; Bitmap is a picture (page 106).

ANSWERS

Chapter 4

Audio and video

A Label the Audacity window

8 marks

1. Edit Toolbar	**2.** Transport Toolbar	**3.** Audio Track	**4.** Tools Toolbar
5. Device Toolbar	**6.** Selection Toolbar	**7.** Mixer Toolbar	**8.** Track Control Panel

1 mark each (page 115).

B Audio or video software?

8 marks

Software	Audio	Video
Wavosaur	☑	☐
Movie Masher	☐	☑
Audacity	☑	☐
AVI Edit	☐	☑
Ardour	☑	☐
Windows Movie Maker	☐	☑
WavePad Sound Editor	☑	☐
Super DVD Video Editor	☐	☑

1 mark each (pages 114 and 118).

C Which command or setting?

8 marks

1. **Add video clips and photos: Home → Add videos and photos**
2. **Add a transition between image frames: the Animations menu**
3. **Set the time an image frame is shown: Animations → Duration**
4. **Make an image frame move: Pan and Zoom**
5. **Add a topic at the start: Home → Title**
6. **Introduce the direction, artists, music and venue: Home → Credits**
7. **Send the clip to a friend by e-mail: For email (File, Save Movie)**
8. **Put the clip on YouTube or Facebook: social media (File, Save Movie)**

1 mark each (pages 120 - 122). Accept the menu word alone (Duration, Title, Credits).

TEACHING TIP

- Install GIMP, Vectorian Giotto, Audacity and Windows Movie Maker before the unit. All four are free.

WRITING SPACES

What Students Write

Read each note out or write it on the board - the students copy it into the space

Chapter 1

No.	Page	What the students write
1	3	**Digital graphics** Digital graphics are drawings or images made with graphic software. Basic elements: pixel, resolution, size and colour.
2	4	**Colours and bits per pixel** Colours = 2 to the power of bpp. 1 bit: 2 colours; 8 bits: 256; 24 bits: 16,777,216 colours.
3	6	**Lossy and lossless** Lossy: smallest file, quality lost (JPEG). Lossless: original quality kept (GIF, PNG, RAW).

Chapter 2

No.	Page	What the students write
4	9	**Save and export** File, Save: an XCF file to edit again. File, Export: a JPEG or PNG file, compressed for its type.
5	9	**Layers** Layers are like transparent sheets: one picture on each layer, so a change does not affect the others.
6	9	**GIMP** GIMP is free, open source image manipulation software. It makes raster graphics, runs on UNIX, Windows and Mac, and saves its files as XCF.

Chapter 3

No.	Page	What the students write
7	12	**Animation** Animation is an optical illusion: frames in sequential order are shown continuously.
8	13	**Types of frames** Key frame: by the user. Tween frames: by the software. Frame: lasts longer. Blank key frame: empty.
9	15	**Uses of animation** Animations are used in commercial publications, to promote products, to develop computer games and to make cartoon films.
10	16	**Frames per second** Smooth motion uses 24 frames per second. With two key frames in 24 frames, the software makes the other 22 tween frames.

TEACHING TIP

- In GIMP, switch on Windows, Single-Window Mode on every computer first. Students lose floating windows.

Chapter 4

No.	Page	What the students write
11	18	**Audacity** Free; records and edits sound. Saves .aup projects; exports WAV, AIFF, FLAC, MP2, MP3.
12	19	**Saving a video** Save Project: a .wlmp file to edit. Save Movie: for e-mail, computer, HD, phones, social media.
13	19	**Audio software** Audacity, Power Sound Editor, Mp3DirectCut, Music Editor Free, Wavosaur, Ardour, WavePad Sound Editor, Sound Engine.

Other pages

No.	Page	What the students write
14	22	**Multimedia in this unit** Graphics with GIMP, 2D animation with Vectorian Giotto, audio with Audacity and video clips with Windows Movie Maker.

MARKING SCHEME

Unit 4 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) colour	**2.** (c) 256
3. (a) dpi	**4.** (a) XCF format
5. (a) an optical illusion	**6.** (b) .swf format
7. (a) Timeline stage	**8.** (c) A and B

2 Fill in the blanks

8 marks (1 each)

1. bitmap	**2.** CMYK	**3.** tertiary	**4.** Channel
5. Brush	**6.** Work Stage	**7.** .aup	**8.** transition

3 Name the pictures

8 marks (1 each)

1. Raster graphic	**2.** Vector graphic	**3.** Crop	**4.** Bucket Fill
5. Linear	**6.** Radial	**7.** Rewind	**8.** Record

Cards 1 - 2: Figure 4.11; 3 - 4: the GIMP toolbox; 5 - 6: the Giotto Brush; 7: the Built-in player; 8: the Audacity Transport Toolbar. Accept Bucket Fill or Paint Bucket for card 4.

4 Answer these

16 marks

1. A graphic uses 16 bits per pixel. How many colours can it show? How many bits per pixel do 32 colours need? (2 marks)

2^{16} = **65,536 colours** (1 mark). 32 = 2^5, so **5 bits** (1 mark).

2. A graphic is 640 pixels wide and 400 pixels high. Find the number of pixels, and its size in bytes with 24 bits per pixel. (2 marks)

640 × 400 = **256,000 pixels** (1 mark). 256,000 × 24 ÷ 8 = **768,000 bytes** (1 mark).

3. Write two differences between lossy and lossless compression. (2 marks)

Any two (1 mark each): lossy reduces the quality, lossless keeps the original quality; a lossy graphic does not reopen in its original form, a lossless one does; lossy gives the smallest files (faster to save and download); JPEG is lossy, GIF / PNG / RAW are lossless.

4. Write two differences between raster graphics and vector graphics. (2 marks)

Any two (1 mark each): raster - an array of pixels, vector - straight or curved lines; raster loses quality when resized, vector does not; raster keeps no records, vector records the lines and colours; raster is not suitable for high quality creations, vector is; raster uses less memory, vector more (Figure 4.11).

5. Write two uses of layers in GIMP. (2 marks)

Any two (1 mark each): each picture goes on its own layer, so a change on one layer does not affect another; the eye shows or hides a layer; the opacity shows the layers below; layers can be moved, duplicated, grouped, anchored, deleted or merged.

6. What is the difference between a key frame and a tween frame? (2 marks)

A key frame is a main static frame that the user creates (e.g. the first and the last frame) (1 mark). Tween frames are created by the software between key frames for a smooth motion (1 mark).

7. Write two functions of Audacity. (2 marks)

Any two (1 mark each): recording live audio; recording music played in the computer; inserting sound effects; editing WAV, AIFF, FLAC, MP2, MP3 or Ogg Vorbis files; copying, trimming, mixing, joining; changing the speed and the pitch.

8. Name two settings of File, Save Movie in Windows Movie Maker and say what each is for. (2 marks)

Any two (1 mark each): For email - to send by e-mail; For computer; For high-definition display; Android Phone - mobile phones; YouTube / Facebook / Vimeo / Flickr - social media; Audio only - only the sound.

TEACHING TIPS

- Let students say the shortcut keys aloud while they click the tools: R rectangle, E ellipse, M move.
- Students should select the part they need before Trim Audio: Trim keeps the selection.

TEXTBOOK MISPRINTS THE WORKBOOK CORRECTS

- p. 84: "250 x 75 pixels" - it is 250 × 175 = 43,750 pixels (the figure says 175).
- p. 83: "bits per pixel = square root of the colours" - use the power of 2 (256 colours = 8 bits).
- p. 86: "000,000,000 to 225,225,225" - 255, 255, 255; "octal numbers" - hexadecimal (F5, 66, 24).
- p. 123: "256 (1 - 256) colours" - 0 - 255; "Triplet colours" - tertiary colours; "Cyen" - Cyan.
- p. 92: the Text tool key "M" (the same as Move) - T. p. 96 / 99: Select, Feather; Filters, Decor.
- pp. 104 - 113: "Vectorian Giatto" - Vectorian Giotto; p. 106: "Liner", "Redial" - Linear, Radial.
- p. 112: the label "volume" points at the dimensions (550 × 400 px), not at a volume.
- p. 117: the Selection Tool is on the Tools Toolbar, not the Transport Toolbar (see the figure).
- p. 122: "Saving an animation" is saving a Movie Maker project (.wlmp).
- p. 88: "Rastor" - raster; p. 104: "Build-in-player" - Built-in player; p. 105: "Insert bimap" - bitmap.

WHAT STUDENTS USUALLY GET WRONG

- They multiply instead of using powers: 8 bits are 2^8 = 256 colours, not 16.
- They forget to divide by 8 when they change bits into bytes.
- They mix up RGB (light, screens) and CMYK (ink, printing).
- They think a vector graphic is made of pixels. Raster = pixels; vector = lines.
- They export from GIMP and forget to save the .xcf, so the layers are lost.
- They think the user draws the tween frames. The software creates them.
- They confuse Save Project (.wlmp) with Save Movie (a video clip to watch and share).

HOMEWORK AND EXTRA PRACTICE

- Work out the colours for 1 to 8 bits per pixel, and the bytes of a 1024 × 768 graphic with 24 bits.
- Find three colour codes in GIMP's colour window and write them as R, G, B and in HTML notation.
- Draw a flip book of 12 small pictures of a moving object; mark the key frames.
- Textbook page 123, items 5 and 6: compare lossy and lossless compression, and raster and vector graphics.

TEACHING TIPS

- Show layers with transparent sheets on an OHP or with plastic files: one drawing on each sheet.
- O/L papers ask for calculations: always ask students to show the multiplication.